



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

November 27, 2025

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Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

$$P(A) = \lim_{n \rightarrow \infty} \frac{k}{n}$$

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

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■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

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$$I_{\nu}(T, \beta) = B_{\nu}(T) (1 - e^{-\tau_{\nu}(\beta)}) \approx B_{\nu}(T) \tau_{\nu}(\beta) = B_{\nu}(T) \Sigma \kappa_{\nu}(\beta) = \frac{2h\Sigma\kappa_0}{c^2\nu_0^{\beta}} \frac{\nu^{3+\beta}}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

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 - suppose we have no prior information on the parameter distribution

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■ *Likelihood:*

$$L(I_v | v, \sigma_v, \theta) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_{v,i}^2}} \exp\left(-\frac{(I_{v,i} - I_v(\theta))^2}{2\sigma_{v,i}^2}\right)$$

Example

■ *Likelihood:*

$$\log L(I_v | v, \sigma_v, \theta) = -\frac{N}{2} \log(2\pi) - \sum_{i=1}^N \sigma_{v,i} - \frac{1}{2} \sum_{i=1}^N \frac{(I_{v,i} - I_v(\theta))^2}{\sigma_{v,i}^2}$$

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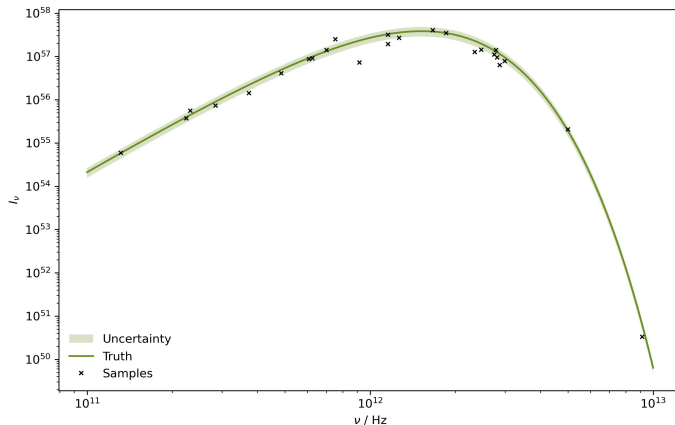
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- *Objective:* minimise the $\chi^2(\theta)$ measure to maximise the posterior probability

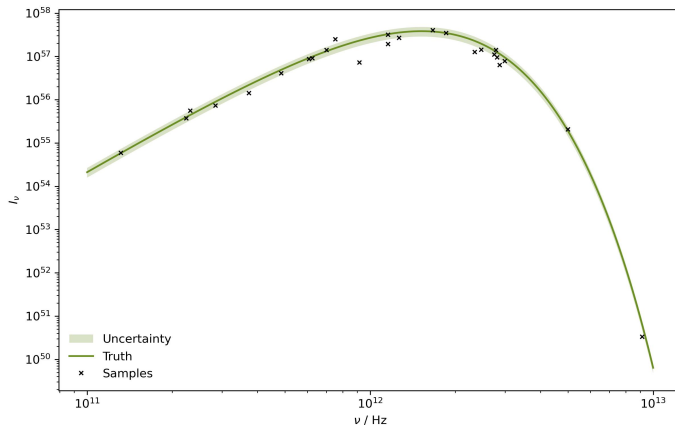
Measurements

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How do we proceed from this dataset to maximise the likelihood for our nonlinear model?



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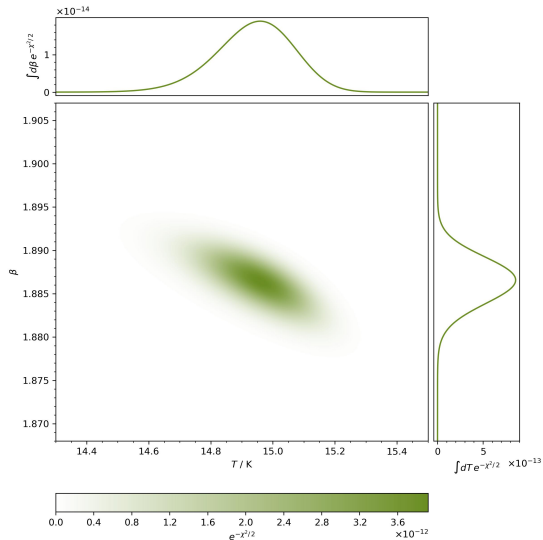
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- *Problem:* as D increases we have N^D gridpoints, and with more complex parameter distributions, this procedure quickly becomes intractable due to computational limits

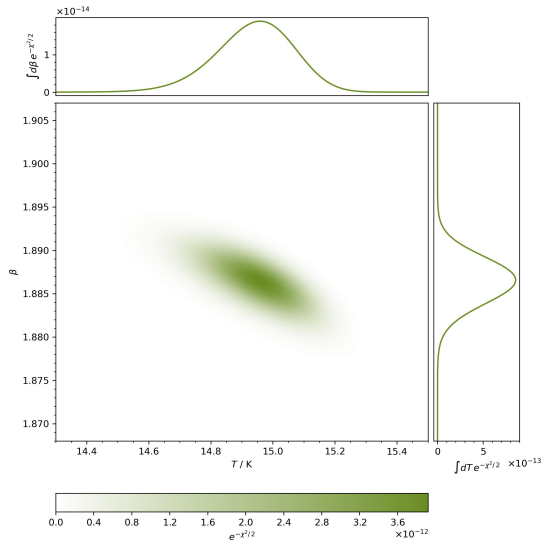
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Can we think of more elegant ways to explore even very high dimensional parameter spaces?



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- *Interactive Visualisations*: check out Ben Frederickson¹ and Andy Casey² to see some nice examples

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 - improve by considering last step and optimizing step range

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 1. initialize at some x_0
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 3. iterate until convergence $|x_{k+1}/x_k - 1| < \epsilon$ or maximum step count is reached

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 - typical settings are $\tau = \kappa = 0.5$ (L. Armijo, *Pac. J. Math.* 1966)

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- *Usage*: found in many modern libraries, suited to specialised use cases like nonlinear least squares, problems with parameter bounds or arbitrary constraints

Application

Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

Limitations & Curses

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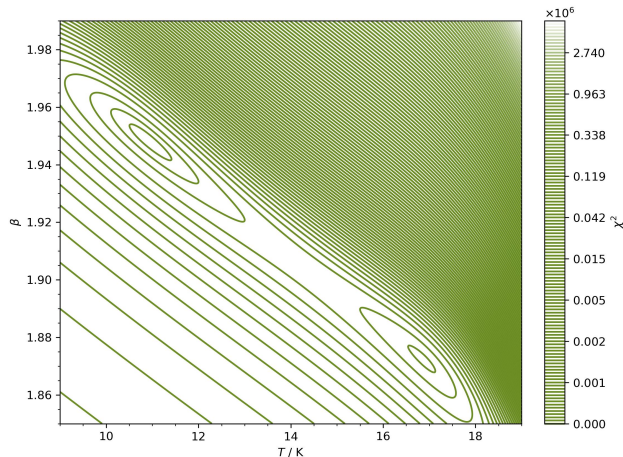
- volume and distance become somewhat meaningless in higher dimensions
- impossible to determine where to step or which parameter is important
- gridding uniformly in linear or logarithmic space is expensive but a useful option
- fixing some parameters or optimising independently are potential solutions

■ *Scaling:*

- parameter scalings of different magnitudes produce pathological stopping criteria

Limitations & Curses

- *Nonconvexity:*
 - convergence does not guarantee global optimum
- *Initialisation:*
 - optimisation can converge to different local optima



Thank You!